FDE Case Study

Lio Procurement Platform

The setup

You've been given the Lio Procurement repository as a folder — you'll find **lio-procurement-app** alongside this brief:

lio-procurement-app

Create your own Git repository from it, create a branch, and implement the tasks below for a customer. When you're done, push it to GitHub, open a pull request against `main`, and treat the description as your hand-off — what you changed, why, and any trade-offs or things you'd do with more time.

Run everything locally with Docker:

`cp .env.example .env` (add your OpenAI API key) → `docker compose up --build`
App → `localhost:3200` · API & docs → `localhost:7200/docs`

Timebox: 5 hours. You're not expected to finish everything — prioritise what delivers the most value, and note what you'd do with more time. If anything is unclear, ask.

The product

Lio is an AI-powered procurement platform. Companies upload vendor-offer PDFs, Lio extracts the details, and each request is tracked through to approval.

- **Multi-tenant** — each *organization* is a separate customer company, fully isolated from the others. The database is seeded with several onboarded companies (Walmart, Schaeffler, Siemens, BMW) plus **Acme Corp**, the customer you'll work with.
- **Roles** — a *requester* creates requests; a *buyer* also approves them; an *admin* does both and manages members, master data, and settings.
- **Master data** — each organization has its own catalog of *suppliers* and *articles* (with negotiated prices).
- **Lifecycle** — a request is created, goes through approval, and moves through status until it is closed.

Sign in to Acme Corp (password `demo1234`):

ACCOUNT	ROLE
`admin@acme.com`	admin

ACCOUNT	ROLE
`buyer@acme.com`	buyer
`requester@acme.com`	requester

Your assignment

Acme Corp's admin has sent over a list of complaints and requests:

1 The PDF extraction isn't reliable

> *"When my team uploads our vendor quotes, the extracted data is often wrong — fields come out incorrect or empty, and sometimes the upload errors out completely. I've attached the five quotes we've been trying it with so you can see for yourself."*

The five quotes are included with this brief.

2 It doesn't use the prices we negotiated

> *"People raise requests and go off buying whatever they find. We've negotiated specific prices for specific articles with our suppliers. When someone creates a request, I want the system to suggest the matching articles we already have on catalog, so we actually use those agreements."*

3 The categorisation doesn't match how we book things

> *"The commodity groups the system assigns don't match our accounting. For us, cable ties always have to be booked under **Electrical Engineering**, printer toner under **IT Hardware**, and small software subscriptions under **Office Supplies** — not where the system keeps putting them. They're quirks on our side, but they're non-negotiable."*

★ Bonus — buyers shouldn't have to approve everything

> *"Requiring a buyer to sign off on every single request wastes too much of their time. Anything below $2,000 shouldn't need approval at all."*

Optional.

Deliverable

No presentation slides needed

Your pull request is the complete deliverable: the code plus the PR description as your hand-off. Please don't prepare slides or a separate presentation.